Casey Barton

Honolulu, HI · (808) 936-5654 · glacier.equilibrium@gmail.com · github.com/GlacierEQ · casey-barton-glaciereq.vercel.app · linkedin.com/in/caseybartonai

PROFILE

Applied AI engineer who shapes how models behave in production: context and prompt engineering, tool/skill/memory design, and self-improvement loops that steer the prompt and improve tool use — built the experiment-driven way, with failure modes found by well-designed research, not vibes. Three years as a founder running an evidence-grade agent control plane: I identify inconsistencies and failure modes in model outputs (stuck loops, bad retrievals, tool misuse, context blowouts), then design the evals and benchmarks that prove the fix. Written rubrics behind every automated quality decision; opinions defended by measured numbers.

The loop is real: **65** authored MCP connectors — governed tool/skill connections with typed contracts — exercised daily by agents whose behavior I tune, measure, and re-tune. No weight fine-tuning claimed; the behavior-shaping surface is eval-driven prompt, context, tool, and memory design.

CORE COMPETENCIES

Model behavior & context engineering: prompt engineering · context engineering · tool/skill/memory design · self-improvement loops · behavioral steering · response-style shaping

Evaluation & research: eval harnesses · golden datasets · behavioral assertions · LLM-as-a-judge with written rubrics · failure-mode research · inconsistency analysis · benchmarks

Agentic AI: domain-specific agents · tool/function calling · MCP · structured outputs · RAG · guardrails · fail-closed recovery · human-in-the-loop gates

Production: Python (primary) · TypeScript · SQL · Bash · REST APIs · JSON Schema / Pydantic-style validation · GitHub Actions (CI/CD) · Postgres · latency & token-cost instrumentation

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present
GlacierEQ — Honolulu, HI

- Shape model behavior across products and use cases: design, test, and optimize the prompts, tools, skills, and memory that determine response quality — context-engineering strategies tuned per domain and model, with the tradeoffs measured.
- Build self-improvement loops: instrument agent runs, find where behavior degrades, steer the prompt and improve tool/skill use and memory recall — the loop compounds instead of resetting every session.
- Run failure-mode research on production outputs: stuck loops, bad retrievals, tool misuse, context blowouts, style inconsistencies — identified through designed experiments, then fixed with behavioral assertions and eval gates that prove the fix held.
- Design the evals that decide what ships: golden datasets, behavioral assertions, LLM-as-a-judge with written rubrics — quality decisions declared up front, measured, and auditable. A **989-test** suite and **62/62** verification checks green back the posture across an **1398-repository** estate.
- Connect behavior to the real world through **65** authored MCP connectors — governed tool/API/data-source connections with typed contracts and structured validation on every boundary.

• Share the intuition: write the context-engineering and prompt-design practices down so every engagement inherits them — playbooks, not tribal knowledge.

EDUCATION

University of Hawaii at Manoa **2016**
Bachelor of Science

AWS Cloud Institute 2025
Cloud Computing program

Colorado State University
Undergraduate coursework

SELECTED PROOF

Behavioral eval infrastructure — quality declared, then measured

• LLM-as-a-judge with written rubrics, golden datasets, behavioral assertions: the exact machinery behind 'identify inconsistencies and failure modes through well-designed research projects.'

Failure posture — designed for production, not the demo

• Stuck loops, bad retrievals, tool misuse, context blowouts: instrumented in production, caught by replay loops and fail-closed gates, recovery restricted to evidence declared before reassessment.

SKILLS

Languages: Python · TypeScript · SQL · Bash · JavaScript / Node.js

Behavior & eval: prompt/context engineering · tool/skill/memory design · self-improvement loops · eval harnesses · LLM-as-a-judge · golden datasets · behavioral assertions · trace intelligence

Production: REST APIs · MCP · Postgres · CI/CD (GitHub Actions) · JSON Schema validation · observability · latency & token-cost instrumentation · retry logic

Evidence boundary: portfolio repositories are independent work. Three years stated as three years — inside the posting's 2–10+ year band, honestly. No weight fine-tuning claimed — behavior shaping is eval-driven prompt, context, tool, and memory design. No Perplexity-internal claims; transferable behavior/eval engineering only. No Go / GCP / GraphQL / K8s / Terraform / DS-statistics depth claimed.